

SCQ (Single Correct Type) :

- If the equation $4y^3 - 8a^2yx^2 - 3ay^2x + 8x^3 = 0$ represent three straight lines, two of them are perpendicular then sum of all possible values of a is equal to :
 (A) $\frac{3}{8}$ (B) $\frac{-3}{4}$ (C) $\frac{1}{4}$ (D) -2
- In a triangle ABC, if A is (1, 2) and the equations of the medians through B and C are $x + y = 5$ and $x = 4$ respectively then B must be :
 (A) (1, 4) (B) (7, -2) (C) (4, 1) (D) (-2, 7)
- The equation of image of pair of lines $y = |x - 1|$ with respect to y-axis is :
 (A) $x^2 - y^2 - 2x + 1 = 0$ (B) $x^2 - y^2 - 4x + 4 = 0$
 (C) $4x^2 - 4x - y^2 + 1 = 0$ (D) $x^2 - y^2 + 2x + 1 = 0$
- If P, Q and R are three points with coordinates (1, 4), (4, 5) and (m, m) respectively, then the value of m for which PR + RQ is minimum, is :
 (A) 4 (B) 3 (C) $\frac{17}{8}$ (D) $\frac{7}{2}$
- The vertices of triangle ABC are A(-1, -7), B(5, 1) and C(1, 4). The equation of the bisector of the angle ABC of $\triangle ABC$ is :
 (A) $y + 2x - 11 = 0$ (B) $x - 7y + 2 = 0$ (C) $y - 2x + 9 = 0$ (D) $y + 7x - 36 = 0$
- Let $f(x)$ be differentiable function on the interval $(0, \infty)$ such that $f(1) = 1$ and

$$\lim_{t \rightarrow x} \left(\frac{t^3 f(x) - x^3 f(t)}{t^2 - x^2} \right) = \frac{1}{2} \forall x > 0$$
, then $f(x)$ is :
 (A) $\frac{1}{4x} + \frac{3x^2}{4}$ (B) $\frac{3}{4x} + \frac{x^3}{4}$ (C) $\frac{1}{4x} + \frac{3x^3}{4}$ (D) $\frac{1}{4x^3} + \frac{3x}{4}$

MCQ (One or more than one correct) :

- The value(s) of t for which the lines $2x + 3y = 5$, $t^2x + ty - 6 = 0$ and $3x - 2y - 1 = 0$ are concurrent, can be :
 (A) $t = 2$ (B) $t = -3$ (C) $t = -2$ (D) $t = 3$

8. If one of the lines given by the equation $ax^2 + 6xy + by^2 = 0$ bisects the angle between the co-ordinate axes, then value of $(a + b)$ can be :
 (A) -6 (B) 3 (C) 6 (D) 12
9. One diagonal of a square is the portion of the line $\sqrt{3}x + y = 2\sqrt{3}$ intercepted by the axes. Then an extremity of the other diagonal is :
 (A) $(1 + \sqrt{3}, \sqrt{3} - 1)$ (B) $(1 + \sqrt{3}, \sqrt{3} + 1)$ (C) $(1 - \sqrt{3}, \sqrt{3} - 1)$ (D) $(1 - \sqrt{3}, \sqrt{3} + 1)$
10. The equation of the sides of the triangle having $(3, -1)$ as a vertex and $x - 4y + 10 = 0$ and $6x + 10y - 59 = 0$ as angle bisector and as median respectively drawn from different vertices, are :
 (A) $6x + 7y - 13 = 0$ (B) $2x + 9y - 65 = 0$
 (C) $18x + 13y - 41 = 0$ (D) $6x - 7y - 25 = 0$
11. If $y = y(x)$ is the solution of differential equation $\frac{dy}{dx} - y = -e^x \tan^2 x$; $y(0)$ then
 (A) $y > 0$ if $x \in \left(0, \frac{\pi}{2}\right)$
 (B) $y < 0$ if $x \in \left(0, \frac{\pi}{2}\right)$
 (C) number of points of inflection for $y = y(x)$ is 0 in $x \in \left(0, \frac{\pi}{2}\right)$
 (D) Tangent drawn to the graph of $y = y(x)$ at $(x, y(x))$ where $x \in \left(0, \frac{\pi}{2}\right)$ lies below the graph of $y = y(x)$.
12. If $M = \{A : A \text{ is a } 3 \times 3 \text{ matrix whose entries are } -1 \text{ and } 1\}$, then
 (A) $|A|$ lies from -6 to 6 (B) $|A| \in \{-4, 0, 4\}$
 (C) $n(M) = 2^9$ (D) $n(M) = 3^9$

Numerical Based Questions :

13. The equation of a line through the mid-point of the sides AB and AD of rhombus ABCD, whose one diagonal is $3x - 4y + 5 = 0$ and one vertex is $A(3, 1)$ is $ax + by + c = 0$. Find the absolute value of $(a + b + c)$ where a, b, c are integers expressed in lowest form.
14. Find the integral value of a for which the point $(-2, a)$ lies in the interior of the triangle formed by the lines $y = x$, $y = -x$ and $2x + 3y = 6$.
15. Let the number of 4×4 matrices whose entries are each '2018' or '-2018' such that the sum of the entries in each row and in each column is '0' are " $15k$ ", then k equals to :

Paragraph Type Questions :

Paragraph for Question Nos. 16 to 17

Consider a triangle ABC with vertex A (2, -4). The internal bisectors of the angles B and C are $x + y = 2$ and $x - 3y = 6$ respectively. Let the two bisectors meet at I.

16. If (a, b) is incentre of the triangle ABC then (a + b) has the value equal to :
(A) 1 (B) 2 (C) 3 (D) 4
17. If (x_1, y_1) and (x_2, y_2) are the co-ordinates of the point B and C respectively, then the value of $(x_1x_2 + y_1y_2)$ is equal to :
(A) 4 (B) 5 (C) 6 (D) 8

Paragraph for Question Nos. 18 to 20

Let $y = f(x)$ satisfy the equation $f(x) = (e^{-x} + e^x) \cos x - 2x - \int_0^x (x-t)f'(t)dt$. Then,

18. $f(x)$ satisfies the differential equation
(A) $\frac{dy}{dx} + y = e^x (\cos x - \sin x) - e^{-x}(\cos x + \sin x)$ (B) $\frac{dy}{dx} = y + e^x(\cos x + \sin x) + e^{-x} (\cos x - \sin x)$
(C) $\frac{dy}{dx} + y = e^x (\cos x + \sin x) - e^{-x}(\cos x + \sin x)$ (D) None of these
19. $f'(0) + f''(0) =$
(A) -1 (B) 2 (C) 0 (D) 1
20. $f(x)$ as a function of x equals
(A) $e^{-x} (\cos x - \sin x) + \frac{e^x}{5} (3\cos x + \sin x) + \frac{2}{5}e^{-x}$
(B) $e^{-x} (\cos x + \sin x) + \frac{e^x}{5} (3\cos x - \sin x) - \frac{2}{5}e^{-x}$
(C) $e^{-x} (\cos x - \sin x) + \frac{e^x}{5} (3\cos x - \sin x) + \frac{2}{5}e^{-x}$
(D) None of these



ANSWERKEY OF CAPS – 15+

1.	(B)	2.	(B)	3.	(D)	4.	(A)	5.	(B)	6.	(C)	7.	(AB)
8.	(AC)	9.	(BC)	10.	(BCD)	11.	(BC)	12.	(BC)	13.	1	14.	3
15.	6	16.	(B)	17.	(D)	18.	(A)	19.	(C)	20.	(C)		